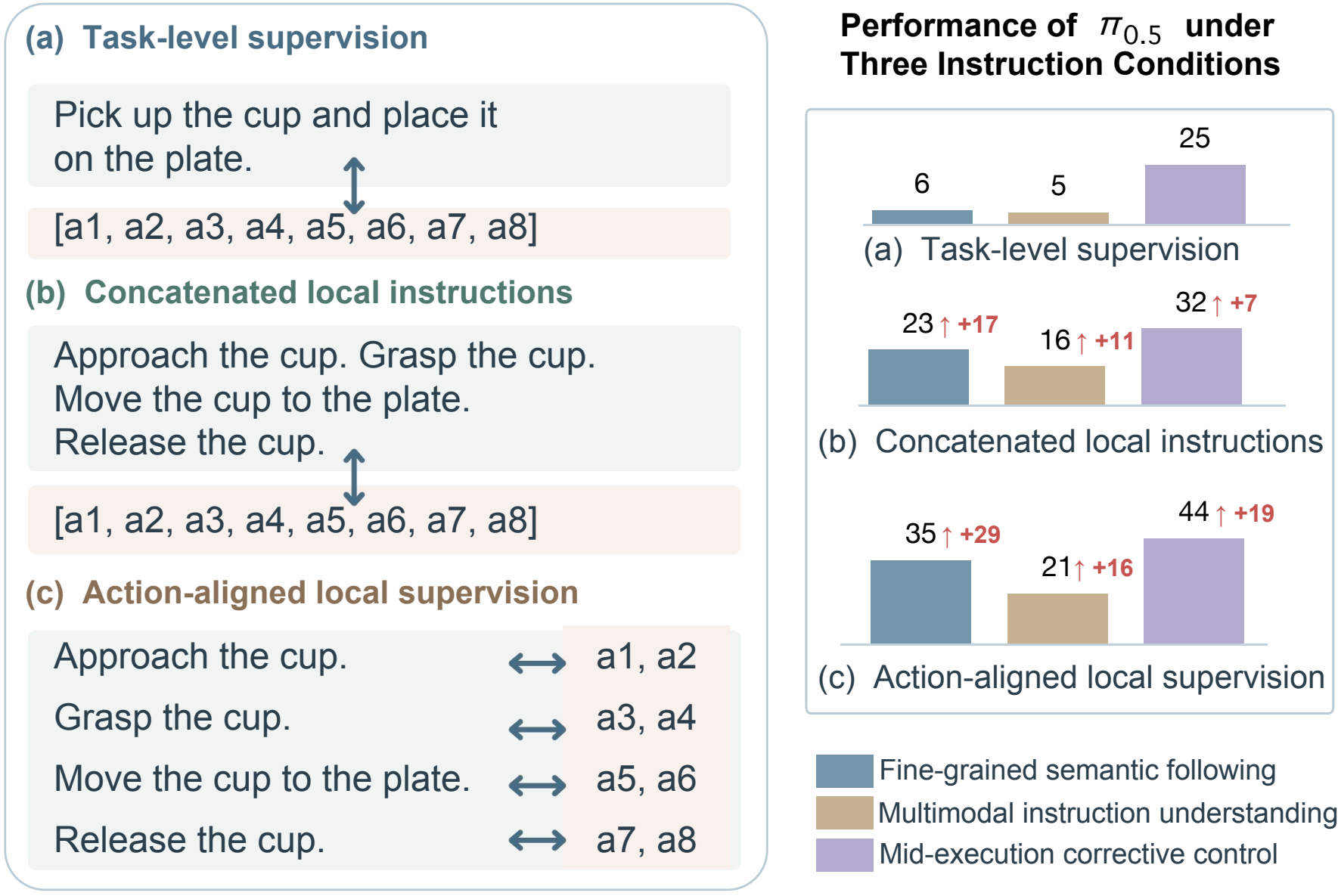

(a) Task-level supervision
Pick up the cup and place it on the plate.
[a1, a2, a3, a4, a5, a6, a7, a8]
(b) Concatenated local instructions
Approach the cup. Grasp the cup.
Move the cup to the plate.
Release the cup.
[a1, a2, a3, a4, a5, a6, a7, a8]
(c) Action-aligned local supervision
Approach the cup. ↔ a1, a2
Grasp the cup. ↔ a3, a4
Move the cup to the plate. ↔ a5, a6
Release the cup. ↔ a7, a8
Performance of $\pi_{0.5}$ under Three Instruction Conditions
6
5
25
(a) Task-level supervision
23 ↑ +17
16 ↑ +11
32 ↑ +7
(b) Concatenated local instructions
35 ↑ +29
21 ↑ +16
44 ↑ +19
(c) Action-aligned local supervision
Fine-grained semantic following
Multimodal instruction understanding
Mid-execution corrective control